

Problem 10.15

Tawny makes a deposit into a bank account which credits interest at a nominal interest rate of 10% per annum, convertible semiannually. At the same time, Fabio deposits \$1,000 into a different bank account, which is credited with simple interest. At the end of 5 years, the forces of interest on the two accounts are equal, and Fabio's account has accumulated to Z . Determine Z .

Solution

Tawny's nominal interest rate is 10% convertible semiannually, so the effective interest rate per half-year is

$$\frac{0.10}{2} = 0.05.$$

The equivalent constant force of interest is

$$\delta_T = 2 \ln(1.05).$$

Thus,

$$\delta_T = 0.09758032834.$$

Let j be Fabio's annual simple interest rate. Under simple interest, the accumulation function is

$$a(t) = 1 + jt.$$

The force of interest at time t is

$$\delta_F(t) = \frac{a'(t)}{a(t)} = \frac{j}{1 + jt}.$$

At time 5, the two forces of interest are equal, so

$$\frac{j}{1 + 5j} = 2 \ln(1.05).$$

Let $\delta = 2 \ln(1.05)$. Then

$$j = \delta(1 + 5j).$$

Therefore,

$$j(1 - 5\delta) = \delta,$$

and

$$j = \frac{\delta}{1 - 5\delta} = 0.1905499730.$$

Fabio's accumulated value after 5 years is

$$Z = 1000(1 + 5j).$$

Substituting the value of j ,

$$Z = 1000(1 + 5(0.1905499730)).$$

Hence,

$$\boxed{Z = 1952.75}.$$